Supersingular poles and signed residues of prime-level Fricke Hauptmoduln

David Jordan

4 November 2026

Abstract

For each of the fifteen prime levels N with $g(X_0(N)^+) = 0$, let $T_N = q^{-1} + O(q)$ be the normalized Hauptmodul for $\Gamma_0(N)^+$. For every prime $p \geq 5$ with $p \nmid N$, we determine the poles and residues of the rational function representing $T_N|U_p$ modulo p. Its poles are exactly the supersingular values with effective ramification index one. If t_i is such a value, its residue multiplied by $\prod_{j\neq i}(t_i - t_j)^{(p+1)/e_j}$ equals $-\left(\frac{-N}{p}\right)$, where the product includes all distinct supersingular values and e_j is the corresponding ramification index. The proof combines a Kaneko–Zagier derivative argument with the explicit evaluation $E_{p+1}^{(p-1)/2} = \left(\frac{6}{p}\right)((p-1)/2)!\,\Delta^{(p^2-1)/24}$ on the supersingular locus. We prove this evaluation uniformly by applying an iterated theta operator to η^{p+1} at full level twelve. It recovers Baker's congruence and, together with Robert's isogeny identity, gives $\left(\Delta(\tau)/\Delta(\ell\tau)\right)^{(p^2-1)/24} = \left(\frac{\ell}{p}\right)$ at every supersingular point of $X_0(\ell)$ for every prime $\ell \neq p$. The ramification indices also admit an interpretation as node thicknesses in the Deligne–Rapoport Fricke quotient. Exact computations provide finite certificates for the five eta-coordinate levels with $p \leq 211$ and for the ten additional levels with $p \leq 47$.

Keywords: supersingular elliptic curves; Fricke groups; Hauptmoduln; Hecke operators; eta quotients; congruences.

1 Introduction

The reduction of U_p applied to a modular function can encode the supersingular locus as a divisor of poles. At level one, this relation is developed by de Shalit (1994) and Kaneko and Zagier (1998, §10); Baker (1998) proves the supersingular congruence that determines the sign in the level-one residue formula. Betina and Lecouturier (2018) obtain corresponding formulas for Legendre modular polynomials. We establish a uniform signed formula on all fifteen prime-level genus-zero Fricke curves. An integral-coordinate lemma removes the need for $X_0(N)$ itself to have genus zero. The factorial evaluation of Theorem 1.3 and the signed discriminant ratio of Theorem 1.5 are independent of this Fricke genus-zero setting.

The residues contain information beyond the support of the supersingular divisor. Since the reduced function vanishes at the cusp and has only simple poles, its residues determine the entire function. An explicit residue formula therefore reconstructs the reduced U_p image from supersingular values and their weights, and detects vanishing when no contributing value remains. The level-one theory provides the model for this reconstruction; the Fricke quotient introduces additional ramification, which must enter both the pole criterion and the product.

The distinction between ramified and unramified supersingular values is essential. A ramified value is removed from the pole divisor, but remains in the residue product with an exponent determined by its ramification index. The algebraic proof makes this distinction through the vanishing order of a quotient involving the Hasse invariant and the derivative of the Hauptmodul. Section 7 explains the same indices as thicknesses of nodes in an integral modular model.

The level-one input is an explicit signed evaluation of $c(E) = E_{p+1}(E)^{(p-1)/2}/\Delta(E)^{(p^2-1)/24}$ at every supersingular curve: it equals $\left(\frac{6}{p}\right)((p-1)/2)!$. The proof applies the Serre–Robert theta-iteration method to $F = \eta^{p+1}$ on $\Gamma(12)$. Euler's pentagonal identity makes the quadratic twist of F explicit, and comparison with the iterated operator selects the sign without a special CM evaluation. Constancy and Baker's squared congruence follow from this formula.

The sign calculation also has a formulation independent of genus zero. Ono (2026, Theorem 1.2 and Remark 4) proves that the supersingular value of $\Delta(\tau)/\Delta(\ell\tau)$ raised to $(p^2-1)/12$ is one, and exhibits examples in which the half-exponent $(p^2-1)/24$ gives -1. Theorem 1.5 determines this remaining sign: it is $\left(\frac{\ell}{p}\right)$ for every prime $\ell \neq p$, at every supersingular point of $X_0(\ell)$. We deduce this discriminant-ratio identity from the factorial evaluation and Robert's isogeny identity. It applies without a genus-zero hypothesis, whereas the explicit residue formula requires the coordinates fixed below.

1.1 Notation and normalization

Throughout the pole and residue results,

$$N \in \mathcal{N} := \{2,3,5,7,11,13,17,19,23,29,31,41,47,59,71\},$$

and $p \geq 5$ is prime with $p \neq N$. These are precisely the prime levels for which $X_0(N)^+$ has genus zero (Ogg, 1975, §3, Corollary). Let T_N denote its unique Hauptmodul with expansion $q^{-1} + O(q)$ at the single cusp. Lemma 2.1 constructs this coordinate over $\Lambda_N := \mathbb{Z}[1/(6N)]$ and proves that it remains a coordinate at every admissible prime. The standard moonshine identification relates these functions to prime-order Fricke McKay–Thompson series (Chen et al., 2019; Conway and Norton, 1979); the proof below uses the modular curves.

For the explicit applications at $N \in \mathcal{N}_0 := \{2,3,5,7,13\}$ we also use

$$a_N = \frac{24}{N-1}, \qquad b_N = N^{12/(N-1)}, \qquad u_N = \left(\frac{\eta(\tau)}{\eta(N\tau)}\right)^{a_N}, \qquad T_N = u_N + a_N + b_N/u_N.$$

The pairs (a_N, b_N) are $(24,4096),(12,729),(6,125),(4,49),(2,13)$. Here u_N is a coordinate on $X_0(N)$ and w_N sends it to b_N/u_N. This eta presentation is not used to define T_N at the ten other levels.

We write $T = T_N$, $q = e^{2\pi i\tau}$, $\theta = q\,d/dq$, $h = (p-1)/2$, $k = (p^2-1)/24$, $A = \overline{E}_{p-1}$ and $B = \overline{E}_{p+1}$. Thus A is the Hasse invariant with expansion 1, and B has the expansion of E_2 modulo p. Put $\Delta = \eta^{24}$ and $\Delta_N = \Delta(\tau)\Delta(N\tau)$. For $f = \sum c_n q^n$, define $f|U_p = \sum c_{pn} q^n$. A bar denotes reduction modulo p. The finite fields are $\mathbb{F}_p$, $\mathbb{F}_{p^2}$ and $\overline{\mathbb{F}}_p$, and $\left(\frac{a}{p}\right)$ is the Legendre symbol. When a modular form is evaluated at an elliptic curve, an invariant differential is understood; weight-zero ratios do not depend on it. The level N and the reduction prime p play distinct roles: p need not belong to $\mathcal{N}$, and $p = N$ is excluded.

1.2 Weights

Let $t_1, \ldots, t_s \in \overline{\mathbb{F}}_p$ be the distinct supersingular values of T. For a point $x = (E, C)$ of $X_0(N)_{\overline{\mathbb{F}}_p}$ over t_i put

$$m_x = |\mathrm{Aut}(E,C)/\{\pm 1\}|, \qquad e_i = m_x\,|\mathrm{Stab}_{\langle w_N\rangle}(x)|.$$

Call t_i *thin* if $e_i = 1$ and put $D_{\mathrm{thin}}(T) = \prod_{e_i=1}(T - t_i)$. For thin t_i let

$$r_i = \mathrm{Res}_{T=t_i} R_{N,p}(T), \qquad W_i = \prod_{j\neq i}(t_i - t_j)^{(p+1)/e_j}.$$

Section 3 shows that e_i does not depend on the representative of the Fricke orbit, that it is the effective (tame) inertia of the quotient map at x (not the order of a group containing ± 1), and that every exponent $(p+1)/e_j$ is an integer.

1.3 Main results

Theorem 1.1. *Let $N \in \mathcal{N}$ and let $p \geq 5$ be a prime with $p \neq N$.*

(a) ***(Rationality.)*** *There is a unique $R_{N,p} \in \mathbb{F}_p(T)$ with $\overline{T|U_p} = R_{N,p}(\overline{T})$. It vanishes at $T = \infty$.*

(b) ***(Pole support.)*** *$R_{N,p} = P/D_{\text{thin}}$ with $P \in \mathbb{F}_p[T]$, $\deg P < \deg D_{\text{thin}}$ and $\gcd(P, D_{\text{thin}}) = 1$. Every pole is simple and has a nonzero residue. In particular $\operatorname{den}_{\text{red}} R_{N,p} = D_{\text{thin}}$.*

(c) ***(Residue as a Hasse derivative.)*** *If t_i is thin and x lies over t_i, then*

$$r_i = \frac{144\,(\theta T)(x)^{p+1}}{B(x)^2}, \qquad \theta = q\frac{d}{dq},$$

a ratio of two mod-p modular forms of weight $2(p+1)$, evaluated at x.

(d) ***(Product.)*** *For thin t_i, with $x = (E, C)$ over t_i and $E' = E/C$,*

$$r_i W_i = \frac{B^{p-1}}{\Delta_N^k}(x) = \left(\frac{N}{p}\right) c(E)\, c(E'), \qquad c(E) := \frac{B(E,\omega)^{(p-1)/2}}{\Delta(E,\omega)^k}.$$

(e) ***(Signed weighted residue formula.)*** *For every thin t_i,*

$$\boxed{r_i W_i = -\left(\frac{-N}{p}\right)} \qquad \text{equivalently} \qquad r_i = -\left(\frac{-N}{p}\right) \prod_{j \neq i} (t_i - t_j)^{-(p+1)/e_j}.$$

In particular $r_i \in \mathbb{F}_p^\times$ and $W_i \in \mathbb{F}_p^\times$, also when $t_i \notin \mathbb{F}_p$.

(f) ***(Hecke form.)*** *With $\Psi_{N,p}(X,Y) = (X^p - Y)(X - Y^p) + pH_{N,p}(X,Y)$ as in Proposition 2.3, $\overline{H}_{N,p}(t^p, t) = \left(\frac{-N}{p}\right) W_i^{-1}$ if $t = t_i$ is thin, and $\overline{H}_{N,p}(t^p, t) = 0$ for every other $t \in \mathbb{F}_{p^2}$.*

Corollary 1.2 (reconstruction)**.** *The supersingular values and their indices determine the entire reduced function:*

$$R_{N,p}(T) = -\left(\frac{-N}{p}\right) \sum_{e_i = 1} \frac{1}{W_i (T - t_i)}. \tag{1.1}$$

Equivalently, if $D_{\text{thin}} = \prod_f f$ is its factorization into monic irreducibles over $\mathbb{F}_p$, then

$$R_{N,p}(T) = \sum_{f \mid D_{\text{thin}}} c_f \frac{f'(T)}{f(T)}, \qquad c_f = -\left(\frac{-N}{p}\right) W_i^{-1} \in \mathbb{F}_p^\times \quad \text{for any root } t_i \text{ of } f. \tag{1.2}$$

Both sums are empty when $D_{\text{thin}} = 1$.

Proof. Subtracting the principal parts leaves a polynomial, which is zero because the function and each summand vanish at infinity. Frobenius-conjugate poles have conjugate residues; the residues lie in $\mathbb{F}_p$ by Theorem 1.1(e), so they are equal on each irreducible factor. The identity $f'/f = \sum_{f(t_i)=0} (T - t_i)^{-1}$ gives the second formula. □

Theorem 1.3 (supersingular factorial evaluation)**.** *For every prime $p \geq 5$, every supersingular elliptic curve $E/\overline{\mathbb{F}}_p$ and every nonzero invariant differential ω,*

$$B(E,\omega)^h = \left(\frac{6}{p}\right) h!\, \Delta(E,\omega)^k. \tag{1.3}$$

Equivalently, in the graded ring $\mathbb{F}_p[E_4, E_6]$,

$$A \mid B^h - \left(\frac{6}{p}\right) h!\, \Delta^k. \tag{1.4}$$

Corollary 1.4 (constancy and Baker's congruence)**.** *The value $c(E)$ is the same for all supersingular $E/\overline{\mathbb{F}}_p$, namely $c_p = \left(\frac{6}{p}\right) h! \in \mathbb{F}_p^\times$. In particular $c_p^2 = -\left(\frac{-1}{p}\right)$, and*

$$E_{p+1}^{p-1} \equiv -\left(\frac{-1}{p}\right) \Delta^{(p^2-1)/12} \pmod{(p, E_{p-1})}.$$

For a prime ℓ, let $Y_0(\ell)$ denote the non-cuspidal modular curve over $\mathbb{Z}[1/\ell]$. For $x = (E, C)$, let $\pi : E \to E' = E/C$ and choose a nonzero invariant differential ω' on E'. Define

$$D_\ell(E, C) := \ell^{12} \frac{\Delta(E, \pi^*\omega')}{\Delta(E', \omega')}. \tag{1.5}$$

The differential weights make this independent of ω' and of the representative of x. It is a unit on $Y_0(\ell)$ over $\mathbb{Z}[1/\ell]$, with q-expansion $\Delta(q)/\Delta(q^\ell)$: on the Tate curve with $C = \mu_\ell$, $\pi^*\omega'_{\mathrm{can}} = \ell\omega_{\mathrm{can}}$. Its complex realization is therefore $\Delta(\tau)/\Delta(\ell\tau)$. We use the same notation for its reduction at $p \neq \ell$.

Theorem 1.5 (signed discriminant ratio)**.** *Let $p \geq 5$ and $\ell \neq p$ be primes. At every supersingular point x of $Y_0(\ell)_{\overline{\mathbb{F}}_p}$,*

$$D_\ell(x)^{(p^2-1)/24} = \left(\frac{\ell}{p}\right).$$

For $\ell = N \in \mathcal{N}_0$, $D_N = u_N^{N-1}$, so this reads $u_N(x)^{(p^2-1)/a_N} = \left(\frac{N}{p}\right)$.

Section 5 proves the factorial evaluation and Corollary 1.4. Section 6 deduces Theorem 1.5 and completes the signed residue formula.

1.4 Consequences

The reconstruction in Corollary 1.2 gives concrete formulas once the supersingular values and their effective indices are known. At the five levels $N \in \mathcal{N}_0$, the connection to Nakaya's squarefree Fricke supersingular polynomials is a change of coordinate: with the normalizations in Nakaya (2019, §4),

$$\mathcal{S}_{N,p}(T) := \prod_i (T - t_i) = ss_p^{(N*)}(T + c_N), \qquad \begin{array}{c|ccccc} N & 2 & 3 & 5 & 7 & 13 \\ \hline c_N & 104 & 42 & 16 & 9 & 0 \end{array} \tag{1.6}$$

For $N = 2, 3, 5, 7$, Nakaya's printed eta formulas have additive constants $128, 54, 22, 13$, giving the shifts displayed above. At level 13 his convention is zero constant term in the q-expansion; this identifies the normalized Hauptmodul with T_{13} and gives $c_{13} = 0$. All constants are reduced modulo p; $\mathcal{S}_{N,p}$ is distinct from the weighted polynomial S of Proposition 3.4. Thus $D_{\mathrm{thin}} = \mathcal{S}_{N,p} / \prod_{e_i > 1} (T - t_i)$, while the omitted values still occur in the weights W_i. For these five levels the exceptional values can also be specified by a polynomial independent of p. Let h_N be the elliptic-value polynomial in Table 2. Lemma 3.2 and Theorem 1.1(b) give

$$D_{\mathrm{thin}}(T) = \frac{\mathcal{S}_{N,p}(T)}{\gcd(\mathcal{S}_{N,p}(T), \overline{h}_N(T))}, \tag{1.7}$$

where the gcd is monic. Thus the denominator is obtained from Nakaya's supersingular polynomial by removing its common factors with the reduction of a fixed characteristic-zero elliptic-value polynomial.

At level two, put $\varepsilon_p = (1 - \left(\frac{-1}{p}\right))/2$ and $\nu_p = (1 - \left(\frac{-2}{p}\right))/2$. The level-two indices described in Section 7 give

$$D_{2,p}(T) = \frac{ss_p^{(2*)}(T + 104)}{(T + 104)^{\varepsilon_p}(T - 152)^{\nu_p}}. \tag{1.8}$$

The two removed values correspond to 0 and 256 in Nakaya's coordinate; they are distinct for $p \geq 5$.

Corollary 1.6 (level-two degree and vanishing)**.** *For every prime $p \geq 5$,*

$$\deg D_{2,p} = \frac{p - 4 + \left(\frac{-1}{p}\right) + 2\left(\frac{-2}{p}\right)}{8}. \tag{1.9}$$

In particular $T_{2A}|U_p \equiv 0 \pmod{p}$ if and only if $p = 5$ or 7.

Proof. The two possible thick values have indices four and two, and supersingularity indicators ε_p and ν_p, respectively. Corollary 3.5 therefore gives $\deg D_{2,p} = (p-1)/8 - \varepsilon_p/4 - \nu_p/2$, which is (1.9). This degree is zero at $p = 5, 7$ and is at least $(p-7)/8 > 0$ for $p \geq 11$. Theorem 1.1(b) proves the last assertion. □

Chen et al. (2019, Lemma 4.8 and Table 4.3) establish p-adic annihilation of T_{2A} at $p = 5, 7$. The corollary specifies the exact first-step modulo-p vanishing criterion. Table 1 illustrates the nonzero cases; at $p = 17$, for example, $R_{2,17} = D'_{2,17}/D_{2,17}$ and both residues are one.

p	$D_{2,p}(T)$	$P_{2,p}(T)$
11	$T - 5$	1
13	$T - 3$	-2
17	$T^2 - T - 8$	$2T - 1$
29	$T^3 - 12T^2 + 12T - 6$	$4T^2 - 5T - 8$

Table 1: Level-two examples, with $R_{2,p} = P_{2,p}/D_{2,p}$. Coefficients use the unique integer representatives in $(-p/2, p/2)$. These entries are included in the certificates of Section 8.

1.5 Relation to earlier work

The derivative argument in Section 4 follows the level-one method presented by Kaneko and Zagier (1998, §10), who trace it to Koike's work on Atkin's congruences (Koike, 1973). The period formula of de Shalit (1994) and the congruence of Baker (1998) supply the level-one precedents for the residue product and its sign; Betina and Lecouturier (2018) treat the corresponding Legendre modular polynomials. The isogeny transformation of B is due to Robert (1980, Théorème B, p. 106; §3.3, Lemma 7 and (19), p. 124); we include a q-expansion proof to fix the differential normalization. Here the level-one factor $-\left(\frac{-1}{p}\right)$ is recovered from Theorem 1.3 and Wilson's theorem.

The factorial evaluation uses a classical mechanism: Serre's functional equation for Eisenstein series, presented by Robert (1980, §2.5, Théorème 2, p. 119), produces factorial coefficients by iterating theta and then evaluating on the supersingular locus. We apply that mechanism to the form η^{p+1} of weight $(p+1)/2$ at full level twelve. Its explicit quadratic twist gives the signed value of c_p simultaneously in all residue classes of p modulo twelve. Robert's cited theorem concerns pairs of even-weight Eisenstein series with complementary weights summing to $p+1$; it does not state the factorial evaluation of Theorem 1.3. The argument here does not use Baker's congruence or the signed discriminant-ratio identity as an input.

Fricke supersingular polynomials are studied by Sakai (2011, 2014), Nakaya (2019) and Morton (2023). These papers treat orthogonal-polynomial representations, factorization and resultant identities. Formula (1.6) identifies the supersingular values used here with that literature; the present residue formula additionally assigns a nonzero coefficient to each thin pole and retains the ramified values in its weighted product.

Jorgenson et al. (2019, §4.2, Table 1 and Appendix C) determine the elliptic values of normalized Fricke Hauptmoduln and their associated class fields. Their j_N equals our T_N. Table 2 records their polynomials for the five eta-coordinate levels. Their Schwarzian equation detects the same local ramification indices that enter our Hasse-derivative calculation; (1.7)

expresses its elliptic-value locus as the locus of possible supersingular pole cancellations. The signed residue formula concerns the remaining thin values.

Duncan and Swisher (2026, Proposition 3.1 and §4) use Deligne's supersingular expansion of $pJ|U_p$ and prime-power level modular functions to characterize the monstrous exponents. Here we determine the signed residues on the fifteen prime-level genus-zero Fricke curves.

Bringmann et al. (2026, Theorem 1.1) study iterated U_p congruences for positive-weight meromorphic cusp forms on $\mathrm{SL}_2(\mathbb{Z})$ whose interior pole reduces to a supersingular point. Their p-adic approximation by holomorphic cusp forms concerns a different setting from our first-step modulo-p pole divisor and signed residue formula.

Landweber (1988, Theorem 3, p. 72; §4) gives the half-exponent identity $\Delta^{(p^2-1)/24} = (e_a - e_b)^{(p^2-1)/4}$ for supersingular models $y^2 = 4\prod_{a=1}^{3}(x - e_a)$ and $a \neq b$. Theorem 1.5 instead compares discriminants along a prime-degree isogeny, with its differential normalization explicit, and determines the sign from Theorem 1.3 and Robert's identity. Ono (2026, Theorem 1.2 and Remark 4) proves the squared isogeny-ratio identity and records examples with nontrivial half-exponent sign. His work develops the supersingular congruences arising in the description of the divided beta family by Behrens (2009, Theorem 1.2) and the explicit calculations of Larson (2020, §5.4, Conjecture 5.17). These connections motivate the discriminant identity; no new statement about the Adams–Novikov spectral sequence is needed here.

Our main result combines the Hasse-derivative calculation with the explicit factorial evaluation to determine the signed residue products on all fifteen prime-level genus-zero Fricke curves. The latter evaluation also gives the discriminant-ratio identity for every prime isogeny degree different from p. The exponent $(p^2 - 1)/24$ and the isogeny transformation of E_{p+1} are classical inputs; the formulas above specify their signed consequences.

2 Standard inputs and the residue criterion

Lemma 2.1 (integral Fricke coordinate)**.** *Let N be prime with $g(X_0(N)^+) = 0$, and put $\Lambda_N = \mathbb{Z}[1/(6N)]$. The coarse quotient*

$$\mathcal{X}_N^+ := X_0(N)_{\Lambda_N}/\langle w_N \rangle$$

is a smooth proper curve over Λ_N with a single cusp section c. There is a unique coordinate T_N over Λ_N whose expansion at that section is $q^{-1} + O(q)$. It identifies $\mathcal{X}_N^+$ with $\mathbb{P}^1_{\Lambda_N}$ and c with infinity. In particular, $T_N \in q^{-1} + q\Lambda_N[[q]]$ and its reduction is a coordinate at every prime $p \geq 5$ with $p \neq N$.

Proof. Away from $6N$ the coarse modular curve $X_0(N)$ is smooth and proper, with geometrically connected fibres; its two disjoint rational cusp sections are exchanged by w_N (Deligne and Rapoport, 1973; Katz and Mazur, 1985). The quotient exists, is proper, and is flat over Λ_N: on invariant affine opens its coordinate ring is a direct summand, via averaging, of a flat Λ_N-algebra. Since 2 is invertible, formation of the quotient commutes with base change. Each geometric fibre is the quotient of a smooth curve by an involution, hence is smooth; thus $\mathcal{X}_N^+$ is smooth over Λ_N. Its fibres are geometrically connected and have constant genus, which is zero by the assumption on the generic fibre. The two cusps give one section c. The quotient map is an isomorphism on the completed neighbourhood of either cusp onto that of c, so $\widehat{\mathcal{O}}_{\mathcal{X}_N^+,c} = \Lambda_N[[q]]$.

Write $f : \mathcal{X}_N^+ \to \operatorname{Spec}\Lambda_N$. On every fibre, $\mathcal{O}(c)$ has degree one and $H^1(\mathcal{O}) = 0$. Cohomology and base change therefore make $f_*\mathcal{O}(c)$ locally free of rank two, and the principal-part sequence is exact:

$$0 \longrightarrow \Lambda_N \longrightarrow H^0(\mathcal{X}_N^+, \mathcal{O}(c)) \longrightarrow \Lambda_N\, q^{-1} \longrightarrow 0.$$

The last identification uses the parameter q to trivialize the normal line of the section. Choose a section with principal part q^{-1} and subtract its constant coefficient. This gives T_N; two such

choices differ by a global constant, proving uniqueness. The sections $1, T_N$ form a basis on every fibre and generate $\mathcal{O}(c)$ there, so they give an isomorphism to $\mathbb{P}^1_{\Lambda_N}$, with the asserted cusp and coordinate. □

The lemma uses genus zero only for the Fricke quotient. Neither its proof nor the arguments below require $X_0(N)$ itself to have genus zero.

We use the q-expansion principle and Kodaira–Spencer isomorphism for prime-to-p level structures (Gross, 1990; Katz, 1973; Katz and Mazur, 1985). On a fine moduli cover, the Hasse invariant has simple zeros precisely at the supersingular points, and Kodaira–Spencer identifies a weight-two form f with $f\,dq/q$. In particular, θT represents dT. We record the statements used repeatedly.

Standard fact I1.

(i) The q-expansion principle holds at prime-to-p level. The forms A and B have q-expansions 1 and E_2 modulo p, respectively.

(ii) The Hasse invariant A has simple zeros precisely at the supersingular points on a fine moduli cover.

(iii) For $p \geq 5$ the stabilizers relevant to $X_0(N)$ and its Fricke quotient are tame, and these coarse quotients commute with reduction. The supersingular values of T lie in $\mathbb{F}_{p^2}$.

(iv) Kodaira–Spencer identifies a weight-two form f with $f\,dq/q$; in particular θT represents dT.

The field-of-definition assertion in (iii) follows by choosing a model with p^2-Frobenius $[-p]$ as follows; every cyclic subgroup of order N is then rational as a subgroup scheme.

Lemma 2.2 (Deuring models)**.** *Let $E_1/\overline{\mathbb{F}}_p$ be supersingular, $p \geq 5$. Then E_1 has a model over $\mathbb{F}_{p^2}$ whose p^2-power Frobenius endomorphism is $[-p]$. For such a model every finite subgroup of order prime to p is $\mathbb{F}_{p^2}$-rational, and E_1/K again has Frobenius $[-p]$.*

Proof. A supersingular curve admits a model over $\mathbb{F}_{p^2}$ (Deuring, 1941; Waterhouse, 1969). For any such model $E/\mathbb{F}_{p^2}$, the p-power Frobenius $F : E \to E^{(p)}$ and the Verschiebung $V : E^{(p)} \to E$ are both purely inseparable of degree p (the Hasse invariant vanishes), so $F^{(p)} : E^{(p)} \to E^{(p^2)} = E$ and V differ by an automorphism u of E: $F^{(p)} = u \circ V$, whence $\pi_{p^2} = F^{(p)} \circ F = u \circ V \circ F = u \circ [p]$. Thus $\pi_{p^2} = p\,u$ with $u \in \mathrm{Aut}(E)$. If $j \neq 0, 1728$ then $u = \pm 1$ and the quadratic twist changes the sign; if $j = 0$ (resp. 1728) the curve $y^2 = x^3 + 1$ (resp. $y^2 = x^3 + x$) over $\mathbb{F}_p$ has trace 0, so $\pi_{p^2} = \pi_p^2 = [-p]$ (Deuring, 1941; Waterhouse, 1969); see also Ono (2026, Lemma 4.3). Since $[-p]$ is central, Galois preserves every subgroup, and the quotient isogeny ϕ satisfies $\pi'\phi = \phi\pi = \phi[-p] = [-p]\phi$, so $\pi' = [-p]$. □

Proposition 2.3. *Let Δ_p be the set of integral matrices of determinant p with lower-left entry divisible by N. There is a unique $\Psi_{N,p} \in \Lambda_N[X, Y]$ with*

$$\Psi_{N,p}(X, T(\tau)) = (X - T(p\tau)) \prod_{b=0}^{p-1} \Big(X - T\big(\tfrac{\tau+b}{p}\big)\Big).$$

It is monic of degree $p+1$ in each variable, symmetric, and satisfies $\Psi_{N,p} \equiv (X^p - Y)(X - Y^p) \pmod{p}$.

Proof. The matrices $\mathrm{diag}(p, 1)$ and $\begin{pmatrix} 1 & b \\ 0 & p \end{pmatrix}$, $0 \leq b < p$, represent $\Gamma_0(N)\backslash\Delta_p$. Conjugation by $W_N = \begin{pmatrix} 0 & -1 \\ N & 0 \end{pmatrix}$ maps $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ to $\begin{pmatrix} d & -c/N \\ -Nb & a \end{pmatrix}$, so it preserves Δ_p. Hence the elementary

symmetric functions of the roots are $\Gamma_0(N)^+$-invariant, holomorphic on the upper half-plane, with a pole only at the single cusp. Their coefficients initially lie in $\Lambda_N[\zeta_p]((q^{1/p}))$. Translation of τ removes fractional powers, and cyclotomic Galois automorphisms permute the roots, so the expansions lie in $\Lambda_N((q))$. Subtracting powers of $T = q^{-1} + O(q)$ then puts them in $\Lambda_N[T]$. Modulo $1-\zeta_p$ all $T((\tau+b)/p)$ coincide with $T(q^{1/p})$, whose p-th power is $T(q)$, and $T(p\tau) \equiv T^p$; this gives the congruence. For irreducibility, the reduction of $\Gamma_0(N)$ modulo p is $\mathrm{SL}_2(\mathbb{F}_p)$, by the Chinese remainder theorem and the surjectivity of reduction for $\mathrm{SL}_2(\mathbb{Z})$. It acts transitively on the $p+1$ cyclic subgroups of order p. The $p+1$ roots displayed above are distinct: their leading terms are q^{-p} and $\zeta_p^{-b}q^{-1/p}$ for $0 \le b < p$. They therefore form one orbit, so $\Psi_{N,p}(X,T)$ is irreducible over $\mathbb{C}(T)$. The pole order of the jth elementary symmetric function is at most $p + (j-1)/p \le p+1$; the constant coefficient has pole order $p+1$. Thus the polynomial has degree $p+1$ in Y as well as in X. For symmetry, $\mathrm{diag}(1,p) \in \Delta_p$ gives $\Psi_{N,p}(T(\tau), T(p\tau)) = 0$ by evaluating the defining relation at $p\tau$. Irreducibility identifies the reversed polynomial with the original up to a rational scalar. The constant coefficient, as a polynomial in X, has leading term $q^{-(p+1)}$ with coefficient one, so $\Psi_{N,p}$ is monic of degree $p+1$ in Y. Comparing the X^{p+1} coefficients in the two polynomials makes that scalar one. Thus $\Psi_{N,p}(Y,X) = \Psi_{N,p}(X,Y)$. □

Define $H = H_{N,p} \in \Lambda_N[X,Y]$ by $\Psi_{N,p} = (X^p - Y)(X - Y^p) + pH$; H is symmetric. Put

$$\psi = \frac{T(q)^p - T(q^p)}{p} \in \Lambda_N((q)),$$

which lies in this ring because $T(q^p) \equiv T(q)^p \pmod{p\Lambda_N}$.

Proposition 2.4 (residue criterion)**.** *In* $\mathbb{F}_p((q))$,

$$\overline{\psi} = \frac{\overline{H}(T^p, T)}{T^{p^2} - T}, \qquad \overline{T|U_p} = \overline{Q}(T) + \overline{\psi},$$

where $\overline{Q} \in \mathbb{F}_p[T]$ *is defined by* $-[X^p]\Psi_{N,p} = Y^p + pQ(Y)$. *Hence* $R_{N,p} = \overline{Q}(T) + \overline{H}(T^p,T)/(T^{p^2} - T)$ *exists and is unique. It vanishes at* ∞*, its poles are simple and lie in* $\mathbb{F}_{p^2}$*, and for* $t \in \mathbb{F}_{p^2}$

$$\mathrm{Res}_{T=t} R_{N,p} = -\overline{H}(t^p, t).$$

Proof. Substitute $X = T(q^p) = T^p - p\psi$ and $Y = T$ into $\Psi_{N,p} = 0$. Since $X - Y^p = -p\psi$, division by p and reduction give $-\overline{\psi}\,(T^{p^2} - T) + \overline{H}(T^p, T) = 0$. The sum of the roots of $\Psi_{N,p}(X,T)$ is $T(q^p) + p\,(T|U_p) = T^p + pQ(T)$, so $T|U_p = Q(T) + \psi$. The polynomial $T^{p^2} - T$ is squarefree, its roots are exactly $\mathbb{F}_{p^2}$, and its derivative is -1. $R_{N,p}$ vanishes at ∞ because $T|U_p \in q\Lambda_N[[q]]$ (the constant term of T is 0). Uniqueness holds because $\overline{T}$ is transcendental over $\mathbb{F}_p$. □

Remark 2.5 (Faber polynomials and replication). The polynomial

$$\mathcal{F}_{N,p}(X) := -[Z^p]\Psi_{N,p}(Z,X) = X^p + pQ(X)$$

is the degree-p Faber polynomial of T: it is the unique monic polynomial for which $\mathcal{F}_{N,p}(T(q)) = q^{-p} + O(q)$. Indeed, the sum-of-roots identity used above gives

$$\mathcal{F}_{N,p}(T(q)) = T(q^p) + p(T|U_p)(q).$$

Under the standard McKay–Thompson identification of T, this is the prime-to-N replication identity; see Conway and Norton (1979, §8) for the replication terminology. The polynomial Q is the correction $(\mathcal{F}_{N,p} - X^p)/p$, rather than the Faber polynomial itself. Proposition 2.4 describes its reduction together with the first Kronecker correction.

Corollary 2.6. *For every* $t \in \mathbb{F}_{p^2}$, $\mathrm{Res}_t R_{N,p} \in \mathbb{F}_p$ *and* $\mathrm{Res}_{t^p} R_{N,p} = \mathrm{Res}_t R_{N,p}$.

Proof. $(\mathrm{Res}_t R)^p = \mathrm{Res}_{t^p} R = -\overline{H}(t^{p^2}, t^p) = -\overline{H}(t, t^p) = -\overline{H}(t^p, t) = \mathrm{Res}_t R$, using $t^{p^2} = t$, the $\mathbb{F}_p$-rationality of R and the symmetry of H. □

Proposition 2.4 gives Theorem 1.1(a) and the a priori bound $\mathrm{den}_{\mathrm{red}} R_{N,p} \mid T^{p^2} - T$ used by the certificates in Section 8.

3 Weights and the weighted supersingular polynomial

Lemma 3.1 (effective inertia)**.** *Let $x = (E, C)$ be a non-cuspidal point of $X_0(N)_{\overline{\mathbb{F}}_p}$, supersingular or ordinary. Let G_x be the group of automorphisms of (E, C) together with the isomorphisms $(E, C) \to w_N(E, C) = (E/C, E[N]/C)$ when $w_N(x) = x$. It is a group once $w_N^2(E, C) \cong (E, C)$ is identified via $[N]$.*

(i) *$|G_x/\{\pm 1\}| = m_x |\operatorname{Stab}_{\langle w_N \rangle}(x)| = e_x$. The value is the same at x and at $w_N(x)$.*

(ii) *$\{\pm 1\}$ is exactly the kernel of the action of G_x on the deformation space of x. Since $e_x \in \{1, 2, 3, 4, 6\}$ is prime to p, $G_x/\{\pm 1\}$ is cyclic and acts faithfully on the tangent line. Hence, for a local parameter z of a fine moduli cover at a point over x, $T - t = z^{e_x} \cdot (\text{unit})$.*

(iii) *If x is supersingular, $(p + 1)/e_x \in \mathbb{Z}$.*

Proof. (i) If $w_N(x) = x$, the isomorphisms $(E, C) \to w_N(E, C)$ form a torsor under $\operatorname{Aut}(E, C)$, so $|G_x| = 2|\operatorname{Aut}(E, C)|$. Conjugation by the isogeny $E \to E/C$ identifies $\operatorname{Aut}(E, C)$ with $\operatorname{Aut}(E/C, E[N]/C)$, so $m_x = m_{w_N(x)}$. (ii) $[-1]$ acts trivially on deformations. A Fricke-type element acts on the deformation functor by $(\widetilde{E}, \widetilde{C}) \mapsto (\widetilde{E}/\widetilde{C}, \widetilde{E}[N]/\widetilde{C})$, followed by the chosen identification of $w_N(E, C)$ with (E, C). If its action were trivial, the universal deformation would admit an isomorphism $\widetilde{E}/\widetilde{C} \cong \widetilde{E}$. Composing with the quotient map would give a degree-N endomorphism of its generic fibre, whose endomorphism ring is $\mathbb{Z}$; this is impossible since N is not a square. Similarly, a nontrivial automorphism other than ± 1 acting trivially would extend to the universal deformation and contradict that generic endomorphism ring. The same argument works at ordinary points. A finite group of order prime to p acting faithfully on $k[[z]]$ is cyclic and acts faithfully on $(z)/(z^2)$, so the invariant ring is $k[[z^e]]$. (iii) $e = 3$ or 6 forces $m_x = 3$, so $j = 0$ is supersingular and $p \equiv 2 \pmod 3$; $e = 4$ forces $m_x = 2$, so $j = 1728$ is supersingular and $p \equiv 3 \pmod 4$. For $p \geq 5$, $m_x \in \{1, 2, 3\}$. □

For explicit calculations at $N \in \mathcal{N}_0$, write $u = u_N$. Then m_x can be read off from the multiplicity μ of $u(x)$ as a root of $P_N(u) - j\, u^N$: $m_x = 3/\mu$ over $j = 0$, $m_x = 2/\mu$ over $j = 1728$, and $m_x = 1$ (with $\mu = 1$) otherwise. Here μ is the ramification index of the coarse map $X_0(N) \to X(1)$. The Fricke stabilizer has order two exactly when $u(x)^2 = b_N$.

Lemma 3.2 (elliptic values and specialization)**.** *Let $N \in \mathcal{N}_0$ and let $h_N(T) = \prod_\tau (T - T_N(\tau))$, where τ runs over the inequivalent elliptic points of $\Gamma_0(N)^+$. These five polynomials are given in Table 2. For every prime $p \geq 5$ with $p \neq N$, its reduction is squarefree, and its roots in $\overline{\mathbb{F}}_p$ are exactly the values with effective index $e > 1$. Consequently,*

$$\gcd(\mathcal{S}_{N,p}, \overline{h}_N) = \prod_{e_i > 1} (T - t_i).$$

Proof. The characteristic-zero values and elliptic orders are those of Jorgenson et al. (2019, §4.2, Table 1 and Appendix C), in the same normalization $T_N = q^{-1} + O(q)$. Their elliptic orders are the effective indices of Lemma 3.1 in characteristic zero. We justify the specialization of their support. A nontrivial effective stabilizer arises either from an automorphism of (E, C) or from a Fricke fixed point. The former can occur only over $j = 0$ or 1728. Since $p \geq 5$ and $p \nmid N$, the corresponding elliptic curve and its automorphisms lift to characteristic zero, and the prime-to-p cyclic subgroup C lifts uniquely over a strict henselian base. An automorphism preserving C on the special fibre therefore preserves its lift. Conversely, these prime-to-p automorphisms persist under reduction. Fricke fixed points are given by $u^2 = b_N$, which is finite étale away from $2N$ and has the same fixed-point locus under specialization. This accounts for all values with $e > 1$ in either characteristic. Directly from the displayed polynomials, the discriminants of h_N, in the order $N = 2, 3, 5, 7, 13$, are

$$2^{16}, \qquad 2^4 \cdot 3^6, \qquad 2^{12} \cdot 5^3, \qquad 2^4 \cdot 3^6 \cdot 7^2, \qquad 2^{12} \cdot 3^6 \cdot 13.$$

Thus distinct elliptic values do not collide at any admissible prime. The roots of $\overline{h}_N$ are precisely the values with $e > 1$; intersecting this set with the supersingular locus proves the gcd formula. □

N	$h_N(T)$	Effective indices
2	$(T+104)(T-152)$	4, 2
3	$(T+42)(T-66)$	6, 2
5	$(T+16)(T^2-12T-464)$	2, 2
7	$(T+10)(T+9)(T-18)$	2, 3, 2
13	$(T+4)(T+3)(T^2-4T-48)$	2, 3, 2

Table 2: Elliptic-value polynomials from Jorgenson et al. (2019). The indices are listed in the order of the displayed factors; an index beside a quadratic factor applies to both roots. At admissible primes, the supersingular roots of $\overline{h}_N$ are exactly the thick values removed in (1.7).

Lemma 3.3. *As mod-p forms on $\Gamma_0(N)$: $A|_{p-1}W_N = \left(\frac{N}{p}\right) A$, $\Delta_N|_{24}W_N = \Delta_N$ and $\theta T|_2 W_N = \theta T$.*

Proof. For a level-one form f of weight k, $f|_k W_N = N^{k/2} f(N\tau)$. The form $E_{p-1}(N\tau)$ has q-expansion $\equiv 1 \equiv E_{p-1}(\tau)$, so it equals A modulo p by Fact I1(i); and $N^{(p-1)/2} \equiv \left(\frac{N}{p}\right)$. The other two identities hold in characteristic 0: $\Delta(-1/(N\tau))\Delta(-1/\tau) = N^{12}\tau^{24}\Delta(N\tau)\Delta(\tau)$, and T is w_N-invariant. □

Proposition 3.4 (weighted supersingular polynomial)**.** *Let $S(T) = \prod_{j=1}^{s}(T - t_j)^{(p+1)/e_j}$. Then $S \in \mathbb{F}_p[T]$ and*

$$S(\overline{T}) = \Delta_N^{-k} = q^{-(N+1)k} \prod_{n \geq 1} (1-q^n)^{-24k}(1-q^{Nn})^{-24k} \quad \text{in } \mathbb{F}_p((q)).$$

Equivalently, $S = A^{p+1}/\Delta_N^k$ as functions on $X_0(N)^+_{\mathbb{F}_p}$. In particular $\sum_j e_j^{-1} = (N+1)(p-1)/24$.

Proof. $F := A^{p+1}/\Delta_N^k$ is a ratio of mod-p forms of weight p^2-1. By Lemma 3.3 it is w_N-invariant, since $\left(\frac{N}{p}\right)^{p+1} = 1$. Δ_N does not vanish on $Y_0(N)$, so F is regular off the cusp of $X_0(N)^+$. Its q-expansion there is $\Delta_N^{-k} = q^{-(N+1)k}(1+O(q))$, because A has q-expansion 1. Hence F is a monic polynomial in T of degree $(N+1)k$. By Fact I1(ii) and Lemma 3.1(ii), F vanishes exactly at the supersingular t_j, to order $(p+1)/e_j$ in $T - t_j$. Comparing degrees gives the mass identity. □

Corollary 3.5. *The degree of the reduced denominator is*

$$\deg D_{\text{thin}} = \frac{(N+1)(p-1)}{24} - \sum_{e_i > 1} \frac{1}{e_i}.$$

In particular $R_{N,p} = 0$ exactly when no thin supersingular value exists.

Proof. Separate the thin and non-thin terms in the mass identity of Proposition 3.4; use Theorem 1.1(b) for the last assertion. □

4 The Kaneko–Zagier step

The following is the level-N version of Kaneko and Zagier (1998, §10, (35)–(38)), who credit the method to Koike.

Lemma 4.1. *In $\mathbb{F}_p((q))$, with $d/dT = (\theta T)^{-1}\theta$,*

$$\frac{d\overline{\psi}}{dT} = T^{p-1} - (\theta T)^{p-1}, \qquad (\theta T)^{p-1} = g(\overline{T}), \quad g := \frac{(\theta T)^{p-1}}{A^2} \in \mathbb{F}_p(T).$$

Proof. Over Λ_N, $\theta\psi = T^{p-1}\theta T - (\theta T)(q^p)$, and $(\theta T)(q^p) \equiv (\theta T)^p$. Divide by $\theta T = -q^{-1} + \cdots$, which is invertible. The function g is a ratio of mod-p forms of weight $2(p-1)$, it is w_N-invariant by Lemma 3.3, and it has q-expansion $(\theta T)^{p-1}$ because $A \equiv 1$. □

Lemma 4.2 (local structure of g)**.** *Let $t \in \mathbb{F}_{p^2}$ be a non-cuspidal point with effective inertia e_t, and put $\varepsilon_t = 1$ if t is supersingular and 0 otherwise. Then*

$$\operatorname{ord}_{T=t} g = \frac{(p-1)(e_t-1) - 2\varepsilon_t}{e_t}.$$

For thin t and x over t, let $a = A/(\theta T)^{(p-1)/2}$, a function on $X_0(N)$ near x. Then a has a simple zero at x,

$$a'(t) := \frac{da}{dT}(x) = \frac{(B/12)(x)}{(\theta T)(x)^{(p+1)/2}} \neq 0, \qquad g = a'(t)^{-2}(T-t)^{-2}(1 + O(T-t)).$$

Proof. By Fact I1(iv) and Lemma 3.1(ii), θT vanishes to order $e_t - 1$ in a stack parameter z, and A to order ε_t; also $T - t \sim z^{e_t}$. This gives the order formula. For thin t, the map $X_0(N) \to X_0(N)^+$ is étale at x and $\theta T(x) \neq 0$. In characteristic 0, for forms F and M of equal weight, $\theta(F/M) = (\partial F \cdot M - F \cdot \partial M)/M^2$, where $\partial = \theta - \frac{k}{12}E_2$ is the Serre derivative. Apply this with $F = E_{p-1}$ and $M = (\theta T)^{(p-1)/2}$ and reduce modulo p; at x we have $A(x) = 0$. The form ∂E_{p-1} has weight $p+1$ and q-expansion $\equiv \theta(1) + \frac{1}{12}E_2 \equiv \frac{1}{12}E_{p+1}$, using the Kummer congruence $2(p+1)/B_{p+1} \equiv 24$ and $\sigma_p \equiv \sigma_1$. So $\partial A = B/12$ as mod-p forms. Finally $da/dT = \theta a/\theta T$. The value $B(x) \neq 0$ because $B/12$ is the derivative of A at its simple zero. □

Proof of Theorem 1.1(b)–(c). Let $t \in \mathbb{F}_{p^2}$. By Proposition 2.4, near t we have $\overline{\psi} = -\overline{H}(t^p, t)(T-t)^{-1} + O(1)$, because $T^{p^2} - T = -(T-t)(1 + O(T-t))$. So the coefficient of $(T-t)^{-2}$ in $d\overline{\psi}/dT$ is $\overline{H}(t^p, t)$. By Lemma 4.1 it is also $-[g]_{-2}$. Hence

$$\operatorname{Res}_t R_{N,p} = -\overline{H}(t^p, t) = [g]_{-2}.$$

By Lemma 4.2, $\operatorname{ord}_t g \geq 0$ unless t is thin. For $p \geq 5$ and $e_t \geq 2$ we have $(p-1)(e_t-1) - 2 \geq 0$. At thin t, $[g]_{-2} = a'(t)^{-2} = 144(\theta T)(x)^{p+1}/B(x)^2 \neq 0$. Every pole is simple by Proposition 2.4, so $R_{N,p} = P/D_{\text{thin}}$ with $P(t_i) \neq 0$ at every thin t_i. Since $R_{N,p}(\infty) = 0$, $\deg P < \deg D_{\text{thin}}$. □

Lemma 4.3 (Robert's isogeny identity)**.** *Let N be any prime $\neq p$ (not only $N \in \mathcal{N}$). Let $x = (E, C)$ be a supersingular point of $X_0(N)$, $\pi : E \to E' = E/C$, and $\omega = \pi^*\omega'$. Then $B(E', \omega') = N\,B(E, \omega)$, $A(E', \omega') = A(E, \omega)$ and $\Delta(N\tau)(x) = N^{-12}\Delta(E', \omega')$.*

Proof. The isogeny identity is Robert (1980, Théorème B; §3.3, Lemma 7); see also Baker (1998, Lemma 2.2). We verify the normalization by q-expansions. For a level-one form f of weight w, $f(N\tau)$ at (E, C, ω) equals $N^{-w}f(E', \omega')$. Check this on $(\mathrm{Tate}(q), \mu_N)$, where $E' \cong \mathrm{Tate}(q^N)$ and $\pi^*\omega'_{\text{can}} = N\omega_{\text{can}}$. The form $E_2(\tau) - NE_2(N\tau)$ is a holomorphic weight-2 form on $\Gamma_0(N)$ with p-integral coefficients. Modulo p, the forms $B(\tau) - NB(N\tau)$ and $A \cdot (E_2 - NE_2(N\tau))$ have weight $p+1$ and the same q-expansion, so they are equal. At a supersingular point this gives $B(E, \omega) = N \cdot N^{-(p+1)}B(E', \omega') = N^{-1}B(E', \omega')$, because $N^{p-1} = 1$. The statement about A follows in the same way from $A(N\tau) = A(\tau)$. □

Proposition 4.4 (weighted residue product)**.** *For thin t_i and $x = (E, C)$ over t_i,*

$$r_i W_i = \frac{B^{p-1}}{\Delta_N^k}(x) = \beta(E) D_N(x)^k = \left(\frac{N}{p}\right) c(E)c(E'), \qquad \beta(E) := \frac{B(E,\omega)^{p-1}}{\Delta(E,\omega)^{2k}} = c(E)^2.$$

Proof. Near x, Proposition 3.4 gives $S(T) = A^{p+1}/\Delta_N^k = a^{p+1}\,Q^k$ with $Q := (\theta T)^{12}/\Delta_N$, a function that is regular and nonzero at x. Hence $W_i = [S/(T-t_i)^{p+1}]_{T=t_i} = a'(t_i)^{p+1}Q(x)^k$. With $r_i = a'(t_i)^{-2}$ this gives

$$r_i W_i = a'(t_i)^{p-1}Q(x)^k = \Big(\frac{B}{12}\Big)^{p-1}(\theta T)^{-\frac{p^2-1}{2}}\cdot\frac{(\theta T)^{\frac{p^2-1}{2}}}{\Delta_N^k} = \frac{B^{p-1}}{\Delta_N^k}(x).$$

The identity $\Delta_N^k = \Delta^{2k}D_N^{-k}$ gives the second equality. For the third, Lemma 4.3 gives $\Delta_N(x) = N^{-12}\Delta(E,\omega)\Delta(E',\omega')$ with $N^{12k} = N^{(p^2-1)/2} = 1$, and $B(E)^{(p-1)/2} = N^{-(p-1)/2}B(E')^{(p-1)/2} = \left(\frac{N}{p}\right)B(E')^{(p-1)/2}$. □

5 The supersingular factorial evaluation

Put $h = (p-1)/2$ and $m = (p+1)/2$, so $m+h = p$ and $m+h(p+1) = pm$. We prove Theorem 1.3 by comparing two expressions for an iterated theta operator applied to $F = \eta^{p+1}$. The leading coefficient calculation is the one underlying Serre's functional equation for Eisenstein series, as presented by Robert (1980, §2.5, Théorème 2, p. 119). The choice of F makes its quadratic twist explicit at every $p \geq 5$.

5.1 A form at full level twelve

Work on a geometrically connected component $X(12)$ of the compactified full-level modular curve, over $R = \mathbb{Z}_{(p)}[\zeta_{12}]$ localized at a prime above p. This is a smooth proper curve carrying the Hodge line bundle ω. Forms of weight w are sections of $\omega^{\otimes w}$; a choice of nonzero invariant differential on an elliptic curve allows evaluation of such a section. The cusp parameter is $q^{1/12}$. We use the integral q-expansion principle on this prime-to-p level (Katz, 1973, §1.6); see also Deligne and Rapoport, 1973.

Lemma 5.1. *The function η^2 is a holomorphic modular form of weight one with trivial character on $\Gamma(12)$. In particular $F = \eta^{p+1} \in M_m(\Gamma(12))$ is integral over R, and*

$$F^{24} = \Delta^{p+1}, \qquad F^{p-1} = \Delta^k. \tag{5.1}$$

Proof. The eta transformation law (Knopp, 1970, Ch. 4) gives $\eta^2(\gamma\tau) = \nu(\gamma)^2(c\tau+d)\eta^2(\tau)$. For $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma(12)$ with $c > 0$, the multiplier formula for d odd (as here, since $12 \mid c$) is

$$\nu(\gamma)^2 = \exp\Big(\frac{2\pi i}{12}\big((a+d)c - bd(c^2-1) + 3d - 3 - 3cd\big)\Big) = 1.$$

Here the squared Kronecker-symbol factor is one, and the displayed numerator is divisible by 12. For $c < 0$, apply the same formula to $-\gamma$: its numerator is -6 modulo 12, so its multiplier squared is -1; the factor $-c\tau - d$ supplies the other minus sign. For $c = 0$, γ is a translation by a multiple of 12 and the assertion follows from $\eta^2(\tau+12) = \eta^2(\tau)$. The form is holomorphic at every cusp. The transformation law and the product expansion give integral cusp expansions over R (the multipliers are twelfth roots of unity), so the q-expansion principle supplies the integral section. Raising to the mth power gives F. Both equalities in (5.1) follow from $\eta^{24} = \Delta$ and $24k = p^2 - 1$. □

5.2 The iterated operator and the twist

Let $\partial_w = \theta - \frac{w}{12}E_2$ be the Serre derivative. It sends p-integral forms of weight w to p-integral forms of weight $w+2$. Define

$$G_0 = F, \qquad G_{i+1} = E_{p-1}\partial_{w_i}G_i + \frac{w_i}{12}E_{p+1}G_i, \qquad w_i = m + i(p+1).$$

Then G_i has weight w_i. The Kummer congruences $E_{p-1} \equiv 1$ and $E_{p+1} \equiv E_2 \pmod p$ show that its q-expansion is $\theta^i F \pmod p$. Moreover

$$G_i = \gamma_i E_{p+1}^i F + E_{p-1} R_i, \qquad \gamma_i = 12^{-i} \prod_{j=0}^{i-1} w_j, \tag{5.2}$$

with R_i a p-integral modular form. This follows by induction: in the recursion the derivative term is multiplied by E_{p-1}, and the remaining term multiplies the leading coefficient by $w_i/12$. At $i = h$, Wilson's theorem gives

$$\overline{\gamma}_h = 12^{-h} \prod_{j=0}^{h-1} (m+j) = 12^{-h} \frac{(p-1)!}{h!} = -\frac{12^{-h}}{h!} \quad \text{in } \mathbb{F}_p. \tag{5.3}$$

Write $\chi_{12}(n) = \left(\frac{12}{n}\right)$ for the Kronecker character, with values $1, -1, 0$ on $n \equiv \pm 1, \pm 5$, and nonunits modulo 12, respectively. Euler's pentagonal identity (Apostol, 1990) takes the form

$$\eta(\tau) = \sum_{n \geq 1} \chi_{12}(n) q^{n^2/24}.$$

Lemma 5.2 (quadratic twist)**.** *The following identity holds between mod-p forms of weight pm on $\Gamma(12)$:*

$$\overline{G}_h = \left(\frac{6}{p}\right) (A^m \overline{F} - \chi_{12}(p) \overline{F}^p). \tag{5.4}$$

Proof. By the product expansion and the freshman's dream, $F \equiv \eta(\tau)\eta(p\tau) \pmod p$. Thus

$$\overline{F} = \sum_{a,b \geq 1} \chi_{12}(a)\chi_{12}(b) q^{(a^2+pb^2)/24}.$$

Applying θ^h multiplies each term by $((a^2+pb^2)/24)^h$, which is zero modulo p when $p \mid a$ and equals $\left(\frac{24}{p}\right) = \left(\frac{6}{p}\right)$ otherwise. The sub-sum with $a = pa'$ is

$$\chi_{12}(p) \sum_{a',b \geq 1} \chi_{12}(a')\chi_{12}(b) q^{p(b^2+pa'^2)/24} = \chi_{12}(p) \overline{F}^p.$$

Consequently the q-expansion of $\overline{G}_h$ equals $\left(\frac{6}{p}\right)(\overline{F} - \chi_{12}(p)\overline{F}^p)$. The integral forms G_h, $E_{p-1}^m F$ and F^p all have weight pm. Since A has q-expansion one, the q-expansion principle proves (5.4) as an identity of sections. □

Proof of Theorem 1.3. Let (E, ω_E) be supersingular and choose a full level-12 structure α with the fixed Weil pairing. Evaluate (5.2) and (5.4) at $x = (E, \alpha, \omega_E)$. Since $A(x) = 0$, they give

$$-\frac{12^{-h}}{h!} B(E, \omega_E)^h \overline{F}(x) = -\left(\frac{6}{p}\right) \chi_{12}(p) \overline{F}(x)^p.$$

Equation (5.1) shows that $\overline{F}(x) \neq 0$ and $\overline{F}(x)^{p-1} = \Delta(E, \omega_E)^k$. Hence

$$B(E, \omega_E)^h = \left(\frac{6}{p}\right) \chi_{12}(p) 12^h h! \, \Delta(E, \omega_E)^k.$$

Euler's criterion gives $12^h = \left(\frac{12}{p}\right) = \chi_{12}(p)$, so the two character factors cancel. This proves the pointwise formula for every supersingular curve, including non-rational j-invariants.

For the divisibility statement, work in $\overline{\mathbb{F}}_p[E_4, E_6]$. The Hasse invariant A is squarefree. Its factors are E_4 if $j = 0$ is supersingular, E_6 if $j = 1728$ is supersingular, and $E_4^3 - j_0 \Delta$ for the

remaining supersingular j_0 (Kaneko and Zagier, 1998, §3). This is the geometric factorization over $\overline{\mathbb{F}}_p$; the irreducible quadratic factors in j over $\mathbb{F}_p$ in Baker (1998, Theorem 1.2) split after extending the constants. For $j_0 \neq 0, 1728$, the factor

$$E_4^3 - j_0\Delta = \left(1 - \frac{j_0}{1728}\right) E_4^3 + \frac{j_0}{1728} E_6^2$$

is irreducible: it has the form $aX^3 + bY^2$ with $ab \neq 0$. The zero set of each factor away from the origin parametrizes the corresponding curves with their nonzero differentials. The pointwise identity therefore forces each factor to divide $B^h - \left(\frac{6}{p}\right) h!\Delta^k$, so their product A divides it. Divisibility descends to $\mathbb{F}_p[E_4, E_6]$. Conversely, divisibility immediately implies the pointwise identity because A vanishes at every supersingular curve. The factors E_4 and E_6 account for the elliptic points $j = 0, 1728$ without extra multiplicities in the graded ring. □

Proof of Corollary 1.4. Theorem 1.3 gives $c(E) = \left(\frac{6}{p}\right) h! \in \mathbb{F}_p^\times$ independently of E. Squaring and using $(h!)^2 = (-1)^{h+1} = -\left(\frac{-1}{p}\right)$ proves the stated form of Baker's congruence. This proof uses neither that congruence nor connectivity of supersingular isogeny graphs as an input. □

Remark 5.3 (a limit to higher-root constancy). Constancy in Corollary 1.4 need not persist for a further square root. For $p \equiv 1 \pmod 8$, the ratio $\rho_p = B^{(p-1)/4}/\Delta^{(p^2-1)/48}$ has weight zero and satisfies $\rho_p^2 = c_p$ at supersingular points, but it need not be constant. For example, modulo 17,

$$A = 8E_4^4 + 10E_4E_6^2, \qquad B = 7E_4^3E_6 + 11E_6^3.$$

The two pairs of form values below have $A = 0$ and $\Delta \neq 0$, and hence represent supersingular elliptic curves with nonzero invariant differentials. Using $\Delta = (E_4^3 - E_6^2)/1728$ gives

j	E_4	E_6	B	Δ	$\rho_{17} = B^4/\Delta^6$
0	0	1	11	3	-2
8	3	3	14	14	2

in $\mathbb{F}_{17}$.

Both values square to $c_{17} = 4$. Thus a uniformly constant quarter-power analogue is false. More generally, whenever $r \mid h$ and $r \mid k$, the ratio $B^{h/r}/\Delta^{k/r}$ takes at most r values, since its rth power is c_p; this bound alone does not imply that all r values occur.

6 The discriminant ratio and signed residues

We use the algebraic unit D_ℓ of (1.5), with $\pi : E \to E' = E/C$ and $\omega = \pi^*\omega'$.

Proof of Theorem 1.5. At a supersingular point $x = (E, C)$, Lemma 4.3 and (1.5) imply

$$\frac{c(E')}{c(E)} = \ell^h \left(\frac{\Delta(E,\omega)}{\Delta(E',\omega')}\right)^k = \left(\frac{\ell}{p}\right) D_\ell(x)^k,$$

because $\ell^{12k} = \ell^{(p^2-1)/2} = 1$. By Corollary 1.4, the left side is one and is nonzero. Therefore $D_\ell(x)^k = \left(\frac{\ell}{p}\right)$. For $\ell = N \in \mathcal{N}_0$, $D_N = u_N^{N-1}$ and $(N-1)k = (p^2-1)/a_N$, giving the assertion for u_N. □

Corollary 6.1 (arbitrary prime-to-p isogenies)**.** *Let $\varphi : E \to E'$ be an isogeny of degree $n \geq 1$ between supersingular elliptic curves over $\overline{\mathbb{F}}_p$, with $(n, p) = 1$. For any nonzero invariant differential ω' on E', put*

$$D_\varphi := n^{12} \frac{\Delta(E, \varphi^*\omega')}{\Delta(E', \omega')}.$$

Then D_φ is independent of ω' and

$$D_\varphi^k = \left(\frac{n}{p}\right).$$

In particular, for cyclic kernels the signed discriminant-ratio identity holds on $Y_0(n)_{\overline{\mathbb{F}}_p}$ for every n prime to p.

Proof. Over $\overline{\mathbb{F}}_p$, factor φ into isogenies of prime degree and apply Lemma 4.3 successively, choosing differentials by pullback. This gives $B(E', \omega') = nB(E, \varphi^*\omega')$. Consequently

$$1 = \frac{c(E')}{c(E)} = n^h \left(\frac{\Delta(E, \varphi^*\omega')}{\Delta(E', \omega')}\right)^k = \left(\frac{n}{p}\right) D_\varphi^k,$$

since $n^{12k} = 1$ in $\mathbb{F}_p$. The Legendre symbol is its own inverse. The weight of Δ gives independence of the chosen differential. □

Remark 6.2 (comparison with Ono). Ono (2026, Theorem 1.2 and Remark 4) uses the reciprocal $R_\ell = \Delta(\ell\tau)/\Delta(\tau) = D_\ell^{-1}$. Since a Legendre symbol equals its inverse, Theorem 1.5 determines the same half-exponent sign for R_ℓ, namely $\left(\frac{\ell}{p}\right)$. For example, $\left(\frac{2}{11}\right) = -1$, as in his example. His Theorem 1.3 states that on a model with p^2-Frobenius $[-p]$ and an $\mathbb{F}_{p^2}$-rational differential, $\Delta(E, \omega)^{2k} = (-1)^{(p+1)/2}$. Theorem 1.3 instead relates $B(E, \omega)^h$ to $\Delta(E, \omega)^k$ for every nonzero differential. The ratio argument requires no descent of a twenty-fourth root of Δ to $\mathbb{F}_{p^2}$.

Proof of Theorem 1.1(e). Proposition 4.4 and Corollary 1.4 give directly

$$r_i W_i = \left(\frac{N}{p}\right) c(E)c(E') = \left(\frac{N}{p}\right) \left(\left(\frac{6}{p}\right) h!\right)^2 = -\left(\frac{-N}{p}\right).$$

By Corollary 2.6, $r_i \in \mathbb{F}_p^\times$, and hence $W_i = -\left(\frac{-N}{p}\right) r_i^{-1} \in \mathbb{F}_p^\times$. □

Proof of Theorem 1.1(f). Proposition 2.4 gives $\overline{H}(t^p, t) = -\operatorname{Res}_t R_{N,p}$. By part (b) this is zero unless $t = t_i$ is thin, in which case it equals $-r_i = \left(\frac{-N}{p}\right) W_i^{-1}$ by part (e). □

Remark 6.3 ($\mathbb{F}_p$-rationality at non-rational points). The two rationality assertions in Theorem 1.1(e) have different explanations. The assertion $r_i \in \mathbb{F}_p$ follows from symmetry of the Hecke polynomial and Frobenius-equivariance of residues (Corollary 2.6). In contrast, writing

$$W_i = \operatorname{Nm}_{\mathbb{F}_{p^2}/\mathbb{F}_p}(D'_{\text{thin}}(t_i)) \prod_{e_j > 1} (t_i - t_j)^{(p+1)/e_j},$$

shows that $W_i \in \mathbb{F}_p$ requires the product of the thick-point factors to lie in $\mathbb{F}_p$. This is not visible factor by factor; it follows here from the signed product identity together with Corollary 2.6: $W_i = -\left(\frac{-N}{p}\right)/r_i \in \mathbb{F}_p^\times$.

7 Geometric interpretation of the weights

The proof of Theorem 1.1 is algebraic. We now interpret its weights in the integral modular curve, retaining the geometric explanation of pole cancellation. Put $W = W(\overline{\mathbb{F}}_p)$ and let

$$\mathcal{B} = X_0(Np)^{\mathrm{DR}}_W, \qquad \mathcal{A} = \mathcal{B}/\langle w_N \rangle.$$

Here $\mathcal{B}$ denotes the coarse Deligne–Rapoport model and w_N is the partial Atkin–Lehner involution. A split node with completed local ring $W[[s, t]]/(st - p^e v)$, v a unit, has *thickness* e.

Proposition 7.1. *The special fibre of $\mathcal{A}$ has two components, each isomorphic to $X_0(N)^+_{\overline{\mathbb{F}}_p}$. They meet at the supersingular Fricke orbits. At the orbit of $x = (E, C_N)$ the node has thickness*

$$e_x = |\operatorname{Aut}(E, C_N)/\{\pm 1\}|\,|\operatorname{Stab}_{\langle w_N\rangle}(x)|.$$

Consequently $R_{N,p}$ has a pole at a supersingular value precisely when the corresponding node has thickness one.

Proof. Deligne–Rapoport's theorem applies with prime-to-p level N and the stabilizer of a line in $(\mathbb{Z}/N)^2$ as the auxiliary group (Deligne and Rapoport, 1973, V, 1.14–1.20; VI, Theorem 6.9, especially (6.9.1)). It gives two copies of $X_0(N)_{\overline{\mathbb{F}}_p}$ with Frobenius gluing, and the coarse completed local ring at x is

$$W[[s,t]]/(st - p^{m_x}), \qquad m_x = |\operatorname{Aut}(E, C_N)/\{\pm 1\}|.$$

The forgetful map $\mathcal{B} \to X_0(N)_W$ is finite: it is proper and has finite fibres on the generic fibre and on the two special-fibre components. Normality identifies $\mathcal{B}$ with the normalization in its generic function field. The generic partial Fricke map covers w_N on $X_0(N)_W$, hence extends uniquely by normalization. On the elliptic locus it is

$$(E, C_N, C_p) \longmapsto (E/C_N, E[N]/C_N, q_N(C_p)).$$

Since q_N is prime to p, it preserves the Frobenius and Verschiebung components and both branches at a node.

To apply Liu's quotient result, note that $\mathcal{B}$ is a normal, proper flat fibred surface over the DVR W, with smooth generic fibre and nodal special fibre. At a fixed geometric node its stabilizer has order two, prime to p, and acts trivially on its residue field $\overline{\mathbb{F}}_p$. As just shown it preserves both branches. It acts nontrivially on either coarse branch. Indeed its action on the smooth curve $X_0(N)$ is a nontrivial tame involution, whose derivative at a fixed point is -1. Taking the branch-preserving quotient therefore multiplies the thickness by this order, by Liu (2002, Proposition 10.3.48(b)–(c), pp. 526–528). At a non-fixed node the inertia is trivial and the thickness is unchanged. Invariants commute with reduction because 2 is invertible in W. The quotient components are consequently $X_0(N)^+_{\overline{\mathbb{F}}_p}$, and no node is smoothed, since no stabilizer exchanges branches. This proves the thickness formula. The last assertion is Theorem 1.1(b). □

Remark 7.2 (level two). For $N = 2$, the Fricke fixed points are $u = \pm 64$, with $T = -104, 152$. At $u = -64$ the reduced automorphism group has order two and the Fricke stabilizer has order two, giving thickness four. At $u = 64$ their orders are one and two, giving thickness two. All other supersingular values have thickness one. The two exceptional values are supersingular precisely when $\left(\frac{-1}{p}\right) = -1$ and $\left(\frac{-2}{p}\right) = -1$, respectively. Thus a thick value is absent from the pole divisor but remains in the weighted residue product. The same distinction applies at all fifteen levels.

8 Finite certificates and computational evidence

The five-level computations below are supplemented by new tests at the other ten levels; their ranges and input models are specified separately. The accompanying exact-arithmetic computations provide checks independent of the signed product formula. They are not used in the proof. We describe the certification bound so that the strength of each computation is explicit.

Proposition 8.1 (finite certificate)**.** *Let $L = p^2$, let $D \in \mathbb{F}_p[T]$ be a monic divisor of $T^L - T$, of degree d, and let $P \in \mathbb{F}_p[T]$ have degree less than d. If*

$$D(\overline{T})\,\overline{T|U_p} - P(\overline{T}) = O(q^{L-d+1}),$$

then $R_{N,p} = P/D$. For $D = 1$, take $P = 0$.

Proof. Proposition 2.4 implies that $R_{N,p} - P/D = C/(T^L - T)$ with $\deg C < L$. If $C \neq 0$, its q-order is $L - \deg C$, since $T = q^{-1} + O(q)$. Multiplication by $D(T)$ gives order $L - \deg C - d \leq L - d$, contradicting the hypothesis. Thus $C = 0$. □

The first computation reconstructs $R_{N,p}$ from its q-expansion, using $D = D_{\text{thin}}$ built independently from supersingular data. The coefficients of the numerator are recovered without the residue formula, and Proposition 8.1 then proves the reconstructed identity. The reduced denominator was also recovered without a candidate for $p \leq 37$. The Hecke polynomial was computed modulo p^2 for $p \leq 31$, checking symmetry, the Kronecker congruence, and the residue criterion at every element of $\mathbb{F}_{p^2}$.

N	certified pairs	thin values	non-rational thin values	range
2	45	530	310	$5 \leq p \leq 211$
3	45	716	456	$5 \leq p \leq 211$
5	44	1059	740	$5 \leq p \leq 211$, $p \neq 5$
7	44	1431	1116	$5 \leq p \leq 211$, $p \neq 7$
13	44	2510	2074	$5 \leq p \leq 211$, $p \neq 13$

Table 3: Certified reconstructions at $N \in \mathcal{N}_0$. Thin values are distinct geometric values of T_N.

All 222 admissible pairs with $p \leq 211$ at these five levels (Table 3) passed the certificates, giving 6246 thin values, of which 4696 are non-rational. Four pairs have no thin values; the ratio set is empty in these cases. In every other pair it is a singleton equal to $-\left(\frac{-N}{p}\right)$. A second calculation, using the double-pole formula of Lemma 4.2, checks a further 960 pairs for $223 \leq p \leq 1499$. That calculation depends on the derivative lemma and is not a separate reconstruction from $T|U_p$.

For each of the ten levels in $\mathcal{N} \setminus \mathcal{N}_0$, the new verification uses the standard elliptic or hyperelliptic model $v^2 = f_N(u)$ of $X_0(N)$ with $w_N(u, v) = (u, -v)$. The stored modular relations have the form

$$j + j \circ w_N = \alpha_N(u), \qquad j\,(j \circ w_N) = Q_N(u), \qquad j = \frac{\alpha_N(u) + \beta_N(u)v}{2},$$

where $Q_N = (\alpha_N^2 - f_N\beta_N^2)/4$. These model data are those used by SageMath's prime-isogeny routines (SageMath Developers, 2026). The coordinate is $T_N = u - s_N$, with $s_N = -[u^{N-1}]\alpha_N/N$. We compute its expansion by formal Newton inversion of $\alpha_N(u(q)) = j(q) + j(q^N)$ and check the product relation separately. The supersingular values are obtained from $j_0^2 - j_0\alpha_N(u) + Q_N(u) = 0$ for supersingular j_0. Indices are determined from the smooth model: $f_N(u) = 0$ detects Fricke ramification, and the multiplicities over $j_0 = 0, 1728$ give the reduced automorphism orders. They are not inferred from the signed residue formula or the weighted Hasse polynomial.

All 122 admissible pairs at these ten levels with $5 \leq p \leq 47$ pass Proposition 8.1. They contain 4056 thin values, of which 3222 lie outside $\mathbb{F}_p$; every weighted residue product has the asserted sign. The weighted Hasse polynomial and mass identity also agree with these independently determined indices. Table 4 gives the counts. A fresh regression run at the original five levels through $p = 47$ passes all 62 admissible pairs. Thus the combined new run covers all 184 admissible pairs at all fifteen levels through $p = 47$, with 4458 thin values, of which 3448 are non-rational.

The eta/Frobenius identity was also tested by a different route: normalized Vélu quotients and their discriminants, without eta functions or Hauptmoduln. For the 15 primes $\ell \leq 47$ and all admissible $5 \leq p \leq 199$, the calculation covers 647 pairs and 127541 cyclic-kernel checks on chosen representatives of supersingular elliptic curves. These are not counts of distinct coarse points: automorphisms may identify kernels over $j = 0$ and 1728. Every kernel gave the asserted discriminant-ratio sign.

N	certified pairs	thin values	non-rational thin values
11	12	136	86
17	12	203	140
19	12	231	170
23	12	267	202
29	12	334	260
31	12	350	272
41	12	445	352
47	12	483	382
59	13	733	614
71	13	874	744
Total	122	4056	3222

Table 4: New certified reconstructions for $5 \le p \le 47$, $p \ne N$, at the ten additional levels.

For Theorem 1.3, the supplementary computations cover all 166 primes $5 \le p \le 997$. Exact polynomial remainders and evaluations on 6430 supersingular j-values, of which 4734 lie outside $\mathbb{F}_p$, agree with the asserted constant. The curve evaluations use three randomly rescaled models per value; additional routes use Legendre and Weierstrass coefficients. Replacing the constant by its negative fails at every prime in this range. The two theta identities used in the proof were also checked as polynomial identities in $\mathbb{F}_p[E_4, E_6]$ at $p = 23, 47, 71, 167$, where $F = \eta^{p+1}$ has level one. These finite checks are independent evidence, not premises of the proof.

Data and software. The supplementary packages `Signed_Weighted_Residue_v1_0` and `Eta_Frobenius_Lemma_Audit_v1_0` contain Python code, per-pair JSON results and reproduction instructions (`python-flint` 0.9.0 and `mpmath`). The package `Factorial_Conjecture_Result_v1_0` contains the factorial-evaluation code and its recorded results under `new_work/`, with reproduction commands in `run_all.sh` (`cypari2` 2.2.4, PARI/GP 2.17.2 and `mpmath` 1.3.0). The package `Fricke_Fifteen_Levels_Extension_v1_0` contains the ten-level verifier, the fresh five-level regression, the model coefficients, source provenance and per-pair certificates (`python-flint` 0.9.0). The reported ranges are those documented by the exact-arithmetic results in these supplementary packages. The code, per-pair data and reproduction instructions are supplied as supplementary files with the manuscript.

9 Further directions

The argument separates an integral genus-zero quotient coordinate, the Hasse divisor and Kodaira–Spencer derivative, and the factorial evaluation of Theorem 1.3. Lemma 2.1 supplies the coordinate uniformly for all fifteen prime Fricke levels; a coordinate on $X_0(N)$ is unnecessary. For composite levels additional cusps and the chosen Atkin–Lehner quotient require separate analysis. For a quotient of positive genus, there is no global Hauptmodul and the residue reconstruction must be reformulated geometrically; neither extension is asserted here.

The proof of Theorem 1.3 suggests seeking other modular forms whose quadratic twists admit an equally explicit expression. Such an extension would require a counterpart of the twist identity, in addition to the standard theta-iteration argument. The genus-zero moonshine identification supplies examples of the modular functions to which the residue theorem applies; the result itself is a statement about their reduction and supersingular divisors.

Acknowledgement of assistance. AI systems assisted in developing arguments, checking calculations and preparing this manuscript, including the factorial-evaluation argument and its

exact-arithmetic verification code. The source computations and proof checks are documented in the supplementary packages. This assistance does not constitute independent specialist refereeing.

References

Apostol, T. M. (1990). *Modular Functions and Dirichlet Series in Number Theory*, 2nd ed. Graduate Texts in Mathematics 41. Springer.

Baker, A. (1998). A supersingular congruence for modular forms. *Acta Arithmetica*, 86(1), 91–100. https://doi.org/10.4064/aa-86-1-91-100.

Behrens, M. (2009). Congruences between modular forms given by the divided β family in homotopy theory. *Geometry & Topology*, 13, 319–357. https://doi.org/10.2140/gt.2009.13.319.

Betina, A., & Lecouturier, E. (2018). Congruence formulas for Legendre modular polynomials. *Journal of Number Theory*, 188, 71–87.

Bringmann, K., Guerzhoy, P., Kane, B., Mertens, M., & Rolen, L. (2026). On U_p-congruences for meromorphic modular forms via supersingularity. *arXiv:2606.14020v1*. https://arxiv.org/abs/2606.14020v1.

Chen, R. C., Marks, S., & Tyler, M. (2019). p-Adic properties of Hauptmoduln with applications to moonshine. *SIGMA*, 15, 033, 35 pp. https://doi.org/10.3842/SIGMA.2019.033.

Conway, J. H., & Norton, S. P. (1979). Monstrous moonshine. *Bulletin of the London Mathematical Society*, 11, 308–339.

Deligne, P., & Rapoport, M. (1973). Les schémas de modules de courbes elliptiques. In *Modular Functions of One Variable II* (LNM 349, pp. 143–316). Springer.

de Shalit, E. (1994). Kronecker's polynomial, supersingular elliptic curves, and p-adic periods of modular curves. *Contemporary Mathematics*, 165, 135–148.

Deuring, M. (1941). Die Typen der Multiplikatorenringe elliptischer Funktionenkörper. *Abhandlungen aus dem Mathematischen Seminar der Hansischen Universität*, 14, 197–272.

Duncan, J. F. R., & Swisher, H. (2026). Modular functions and the monstrous exponents. *arXiv:2602.09135v1*, 9 February 2026. https://arxiv.org/abs/2602.09135v1.

Gross, B. H. (1990). A tameness criterion for Galois representations associated to modular forms (mod p). *Duke Mathematical Journal*, 61(2), 445–517.

Jorgenson, J., Smajlović, L., & Then, H. (2019). The Hauptmodul at elliptic points of certain arithmetic groups. *Journal of Number Theory*, 204, 661–682. https://doi.org/10.1016/j.jnt.2019.03.021. The elliptic-value tables used here are also available in *arXiv:1602.07426v2*, §4.2, Table 1 and Appendix C. https://arxiv.org/abs/1602.07426v2.

Kaneko, M., & Zagier, D. (1998). Supersingular j-invariants, hypergeometric series, and Atkin's orthogonal polynomials. In *Computational Perspectives on Number Theory* (AMS/IP Studies in Advanced Mathematics 7, pp. 97–126).

Katz, N. M. (1973). p-adic properties of modular schemes and modular forms. In *Modular Functions of One Variable III* (LNM 350, pp. 69–190). Springer.

Katz, N. M., & Mazur, B. (1985). *Arithmetic Moduli of Elliptic Curves* (Annals of Mathematics Studies 108). Princeton University Press.

Knopp, M. I. (1970). *Modular Functions in Analytic Number Theory*. Markham.

Koike, M. (1973). Congruences between modular forms and functions and applications to the conjecture of Atkin. *Journal of the Faculty of Science, University of Tokyo, Section IA, Mathematics*, 20, 129–169.

Landweber, P. S. (1988). Supersingular elliptic curves and congruences for Legendre polynomials. In *Elliptic Curves and Modular Forms in Algebraic Topology*, Lecture Notes in Mathematics 1326, pp. 69–93. Springer.

Larson, D. M. (2020). Explicit modular forms from the divided beta family. *Topology and its Applications*, 273, 107094. https://doi.org/10.1016/j.topol.2020.107094.

Liu, Q. (2002). *Algebraic Geometry and Arithmetic Curves*. Oxford Graduate Texts in Mathematics 6. Oxford University Press.

Morton, P. (2023). Supersingular conjectures for the Fricke group. *International Journal of Number Theory*, 19(4), 803–828. https://doi.org/10.1142/S1793042123500409; preprint https://arxiv.org/abs/2105.14532.

Nakaya, T. (2019). The number of linear factors of supersingular polynomials and sporadic simple groups. *Journal of Number Theory*, 204, 471–496. https://doi.org/10.1016/j.jnt.2019.04.015.

Ogg, A. P. (1975). Automorphismes de courbes modulaires. *Séminaire Delange–Pisot–Poitou, Théorie des nombres*, 16 (1974–1975), no. 1, exposé 7, 1–8. https://www.numdam.org/item/SDPP_1974-1975__16_1_A4_0/.

Ono, K. (2026). Modular forms for chromatic homotopy: supersingular congruences. *arXiv:2509.16175v2*, 3 July 2026. https://arxiv.org/abs/2509.16175v2.

Robert, G. (1980). Congruences entre séries d'Eisenstein, dans le cas supersingulier. *Inventiones Mathematicae*, 61, 103–158.

The SageMath Developers (2026). *Isogenies of small prime degree*, including `_hyperelliptic_isogeny_data`, in the SageMath source code. https://github.com/sagemath/sage/blob/develop/src/sage/schemes/elliptic_curves/isogeny_small_degree.py. The exact coefficient data and source checksum used here are preserved in the supplementary extension package.

Sakai, Y. (2011). The Atkin orthogonal polynomials for the low-level Fricke groups and their application. *International Journal of Number Theory*, 7(6), 1637–1661. https://doi.org/10.1142/S1793042111004460.

Sakai, Y. (2014). The Atkin orthogonal polynomials for the Fricke groups of levels 5 and 7. *International Journal of Number Theory*, 10(8), 2243–2255. https://doi.org/10.1142/S1793042114500766.

Waterhouse, W. C. (1969). Abelian varieties over finite fields. *Annales Scientifiques de l'École Normale Supérieure (4)*, 2, 521–560.

Maynooth University Social Sciences Institute (MUSSI), Maynooth University, Maynooth, County Kildare, Ireland
Email address: David.Jordan@mu.ie